Aerial Data Processing-Creating Orthomosaics and DEMs without GCPs

1.0 Introduction to Aerial Photogrammetry

The process for creating high-resolution aerial maps—specifically orthomosaics and Digital Elevation Models (DEMs)—using the Agisoft Metashape software. The guide focuses on a modern workflow that does not require Ground Control Points (GCPs), relying instead on high-precision location data captured directly by an advanced drone system. This approach dramatically reduces field survey time and logistical complexity, making high-accuracy mapping more accessible. The following sections will first define the core products of this process before detailing the step-by-step workflow.

2.0 Understanding the Core Products: Orthomosaics and DEMs

Before detailing the processing workflow, it is essential to understand the fundamental outputs of aerial data processing. An orthomosaic and a Digital Elevation Model are not simply images; they are powerful, data-rich products that serve as invaluable professional tools across numerous industries. Clarifying what they are and why they are valuable provides the necessary context for the subsequent processing steps.

2.1 Defining the Orthomosaic

An orthomosaic is a geo-referenced, distortion-free image created by stitching and correcting numerous individual aerial photos. It can be thought of as a custom, high-resolution, and up-to-date "sharper version of Google Earth" for a specific area of interest.

Also referred to as an "orthophoto," an orthomosaic's primary value comes from the meticulous photogrammetric process that mathematically removes perspective distortions. These distortions, caused by factors like camera tilt, terrain relief, and the physical characteristics of the camera lens, are corrected to achieve an image with two critical properties:

- **No Distortion:** Every point in the image is shown as if viewed from directly above.
- **Uniform Scale:** The image has a consistent scale across its entire extent, allowing for accurate distance and area measurements directly from the map.

2.2 Defining the Digital Elevation Model (DEM)

A Digital Elevation Model (DEM) is a raster grid representation of a surface. In a DEM, each pixel in the grid does not store color information, but rather a specific elevation value. It effectively represents the height of the terrain at any given X,Y coordinate location, providing a detailed three-dimensional understanding of the landscape's topography.

The creation of both orthomosaics and DEMs with high absolute accuracy has traditionally been governed by the use of Ground Control Points.

3.0 The Critical Role of Ground Control Points (GCPs)

The absolute accuracy of aerial mapping products is paramount for professional applications where measurements must correspond precisely to the real world. Ground Control Points (GCPs) have traditionally served as the foundation for ensuring this accuracy, acting as the critical link that anchors the digital model to its true position on Earth.

GCPs are **surveyed physical points on the ground** with precisely measured, known X, Y, and Z coordinates. In the photogrammetric process, their function is to "directly control geo-referencing and accuracy" of both the DEM and the orthomosaic. During a processing step known as "bundle adjustment," GCPs are used to refine the calculated camera positions and internal camera parameters. This ensures that the final 3D model matches the real-world coordinates with a high degree of fidelity.

The primary consequence of omitting GCPs from a traditional workflow is significant. As one key insight notes, **"without GCP, the model is only relative."** This means that while the model may be internally consistent and dimensionally correct, it is not accurately locked into a real-world coordinate system. For example, the dimensions of a building on the map might be correct relative to each other, but the entire building could be located 10 meters north of its actual position on Earth.

The ideal accuracy chain when using GCPs can be summarized as follows:

- GCP → Accurate camera positions
- Accurate point cloud
- Accurate DEM → Accurate orthomosaic

The workflow detailed next, however, utilizes an alternative high-precision data collection method (PPK) that makes a GCP-less process not only possible but highly effective.

4.0 Step-by-Step Workflow in Agisoft Metashape (Without GCPs)

This section provides a detailed, step-by-step guide to processing aerial data in Agisoft Metashape Professional. The workflow is based on a project that does not use Ground Control Points. The source data for this tutorial was acquired with a **WingtraOne drone** using a **precise PPK method**, a key technological factor that provides high-accuracy camera location data and enables this workflow's success.

4.1 Step 1: Add Photos

The initial step is to import the aerial images into the project by selecting either Add Photos... or Add Folder... from the Workflow menu. Once loaded, the initial camera positions will be displayed in the Model window.

Camera reference coordinates can then be loaded using one of three methods on the Reference pane:

1. **Automatic Loading from EXIF:** The software can automatically import standard GPS data directly from the image files' EXIF tags upon import.
2. **Manual Trigger from EXIF:** If not loaded automatically, you can manually trigger the import by clicking the Import Reference from the EXIF button on the Reference pane.
3. **Import from a CSV File:** For high-precision datasets, camera coordinates can be imported from a text file. Click the Import Reference button, browse to your file, and in the Import CSV dialog, specify parameters like the coordinate system, delimiter, and column assignments. Accuracy data can also be loaded from the CSV or entered manually for a specific image by right-clicking it and selecting the Modify... command.

4.2 Step 2: Align Photos

With the photos and their positions loaded, the next step is photo alignment. This is initiated by selecting Align Photos... from the Workflow menu. A dialog box will appear to set reconstruction parameters. After clicking OK, the software analyzes the photos to find common features, or tie points. Upon completion, the refined camera positions and a sparse tie point cloud become visible in the Model view.

4.3 Step 3: Optimize Camera Alignment

This optimization process improves alignment and reduces errors. To begin, click the Optimize toolbar button on the Reference pane. In the Optimize Camera Alignment dialog box that appears, you can select which parameters to refine. For datasets captured with RTK/PPK methods where no GCPs are used, the Fit additional corrections option is particularly beneficial for achieving better overall accuracy. Click OK to start the optimization.

4.4 Step 4: Build Dense Point Cloud

Using the aligned photos and optimized camera positions, a dense point cloud is generated. This is accomplished by selecting Build Point Cloud... from the Workflow menu. In the subsequent dialog box, select the desired quality and filtering parameters, then click OK. This step reconstructs the detailed 3D structure of the surveyed area, resulting in millions of precisely located points.

4.5 Step 5: Build Digital Elevation Model (DEM)

From the dense point cloud, the Digital Elevation Model is created using the Build DEM... command from the Workflow menu. The DEM is calculated for the area defined within the project's bounding box. In the Build DEM dialog box, set the desired parameters for the model and click OK. This process interpolates the point cloud data into a continuous raster grid.

4.6 Step 6: Build Orthomosaic

The final model generation step is to build the orthomosaic. Select the Build Orthomosaic... command from the Workflow menu. A dialog box will appear where you can set the parameters for the final image. After clicking OK, this process projects the original photos onto the DEM, stitching them together and correcting for all distortions to create a seamless, map-accurate image.

With the core map products generated, the final step is to export them for practical use.

5.0 Exporting Final Results

The final and most crucial step is exporting the generated DEM and orthomosaic into standard file formats. This allows the products to be used in Geographic Information System (GIS) applications, Google Earth, CAD software, and other professional platforms for analysis, measurement, and visualization.

5.1 Exporting the DEM

The process for exporting the Digital Elevation Model is straightforward:

1. Select Export DEM... from the File menu.
2. Specify the desired output coordinate system.
3. Optionally, check Write KML file (for use in Google Earth) or Write World file (for georeferencing in GIS applications) to create the necessary supporting files.
4. Select the destination folder, choose the file type, provide a file name, and then save.

Metashape also offers the ability to export the DEM with a false-color palette based on height information, which is useful for visualization. This is supported for TIFF, JPEG, JPEG2000, and PNG formats.

Supported DEM Formats:

- GeoTIFF elevation data (*.tif)
- Arc/Info ASCII Grid (*.bil)
- XYZ file format (*.xyz)
- Sputnik KMZ (*.kmz)
- Google Map Tiles (*.zip)
- MBTiles (*.mbtiles)
- World Wind Tiles (*.zip)
- Tile Map Service Tiles (*.zip)

5.2 Exporting the Orthomosaic

Exporting the final orthomosaic follows a similar procedure:

1. Select Export Orthomosaic... from the File menu.
2. Specify the output coordinate system for the final image.
3. Optionally, check Write KML file or Write World file to generate the georeferencing files needed for Google Earth or GIS software.
4. Select the destination folder, choose the desired file type, provide a file name, and then save.

Supported Orthomosaic Formats:

- TIFF/GeoTIFF (*.tiff)
- JPEG2000 (*.jp2)
- JPEG (*.jpg)
- PNG (*.png)
- BMP (*.bmp)
- Google Earth KMZ (*.zip)
- MBTiles (*.mbtiles)
- World Wind Tiles (*.zip)
- Tile Map Service Tiles (*.zip)

Following these steps completes the entire workflow, successfully transforming a set of raw aerial photos into tangible, accurate, and usable digital map products ready for professional application.